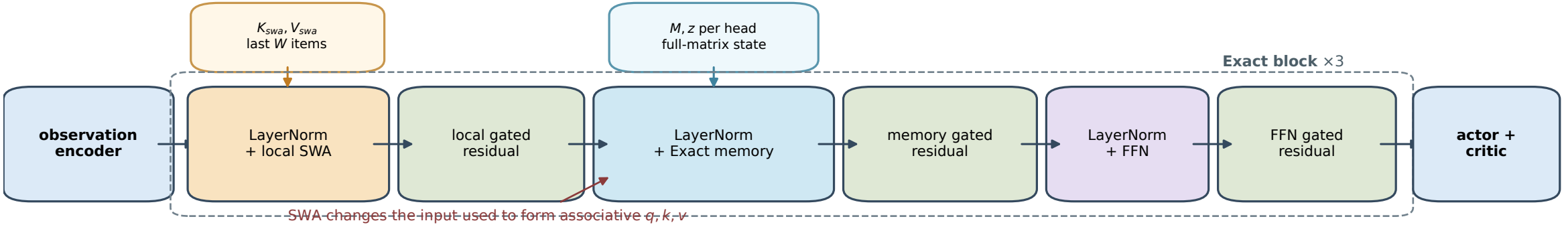
SdLG-Exact reference architecture

Local and associative processing are sequential within every block



Architectural variants and study controls

Positional phase

RoPE rotates Exact q, k before φ; NoRoPE leaves them unchanged.

Approximate operation

Replace (M, z) and its read with $\widetilde{K}, \widetilde{V}, \theta$ and a factorized read.

Original factorial

Also changes layer placement and routes the attenuated trunk to both heads.

Matched control

Restores three complete blocks and one unattenuated shared trunk.

Memory-update gates (β, γ), residual gates, and actor-gradient scale (α) act at different levels.